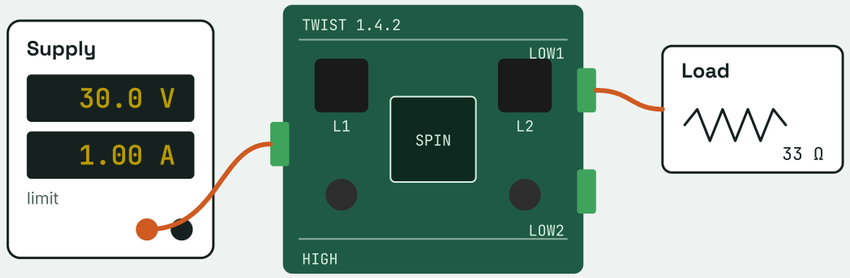
OWNSTUDIO · DESIGN BOARD · ROUND 3

How the setup could look on the canvas

Four ways to draw the same setup: a 30 V supply, the Twist running leg 1 as a buck, and a 33 Ω load on LOW1. Then the inside of the Twist, and how the simulation coverage can be shown without a table. All drawings show the same values so you compare styles, not content.

The outside: source, Twist, load

A Front panels

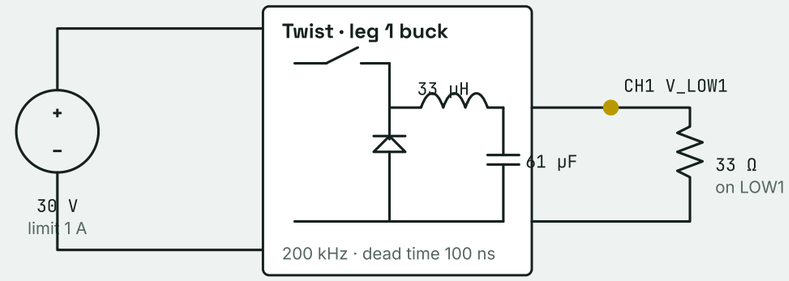


Wires are power; their colour follows the safety envelope.

Close to: Typhoon HIL SCADA panels, LabVIEW front panels, a photo of the bench.

+ Reads like the bench. The Lab checklist (HIGH 30.0 V measured) has an obvious place: the supply's display.
− Each instrument needs its own artwork, and it says nothing about the circuit inside.

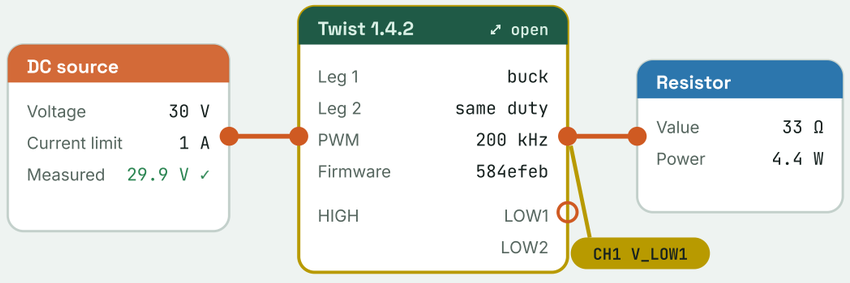
B Schematic



Close to: PLECS, PulsimGui, Simulink with Simscape, LTspice.

+ Engineers read it at once; values sit on the components, and probes land on a node.
− Invites "let me edit the circuit". We'd be one step from a schematic editor, which you ruled out with PulsimGui (Q5).

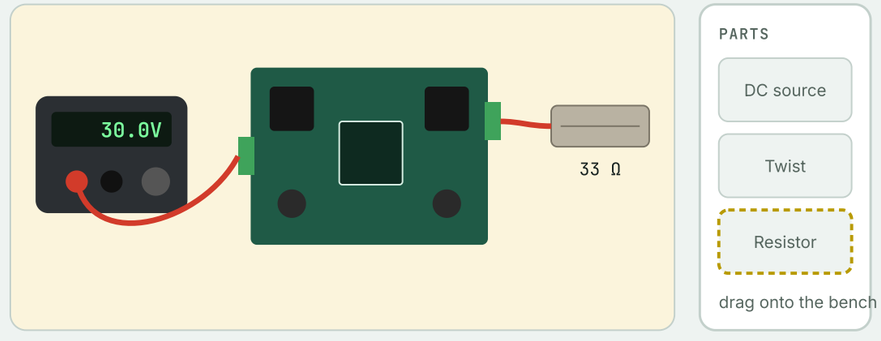
C Nodes with typed ports



A power port only accepts a power wire: source → HIGH and LOW → load, nothing else snaps.

Close to: Node-RED, Unreal Blueprints, Simulink subsystems, React Flow apps.

D Illustrated bench, parts dragged in



Close to: Wokwi, Tinkercad Circuits, Fritzing's breadboard view.

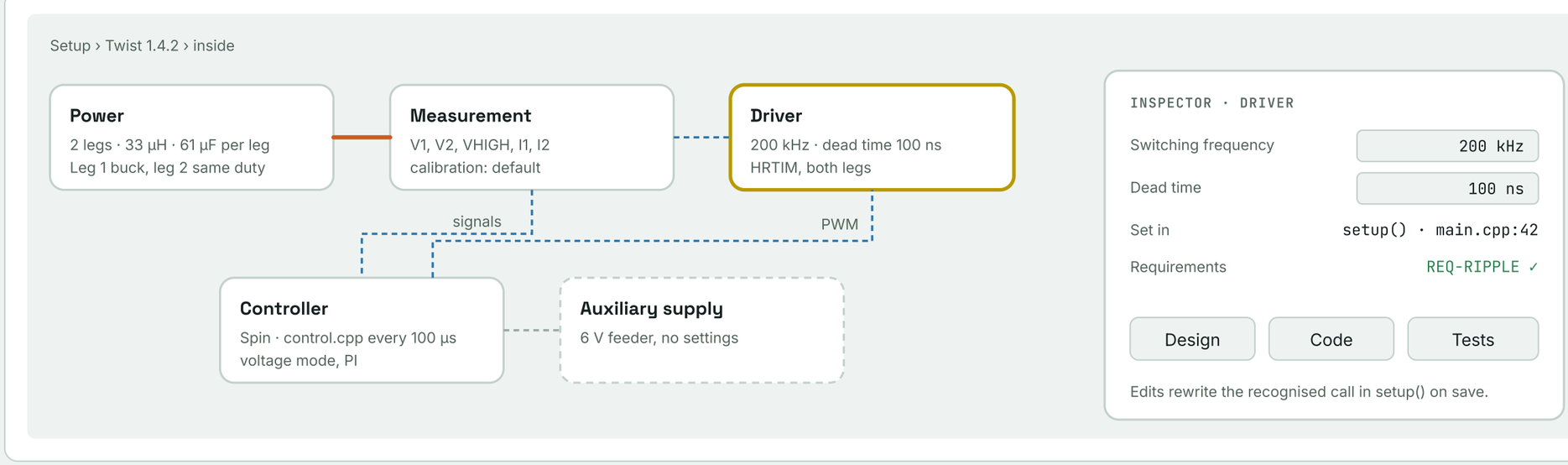
+ Settings live on the node, so the inspector is optional. Typed ports make your rule (no source–Twist–source) visible: the wire won't snap.
− Generic look; less "power electronics" at first sight.

+ The friendliest for your drag-and-drop first steps; a colleague recognises their bench.
− The most artwork per part (every board version, every instrument), and settings need the inspector.

My recommendation: C for the canvas, with D's palette. Nodes carry their values and their live state, typed ports enforce the topology rules, and the look scales to new boards without new artwork. The Twist node gets a small board picture in its header so it still reads as hardware. B stays for the read-only "simulator window" if we ever draw one.

The inside of the Twist

Double-click the Twist node, or "open". The five blocks keep the arrows of today's Home diagram, drawn as nodes. Clicking a block fills the inspector on the right panel with its settings and the three buttons you asked for.



What the simulation covers, without a table

Your Q6: the simulation sheet shown on the canvas itself, in its Simulate tint. Each block gets one fidelity chip, the way Simscape blocks offer a modelling fidelity and PLECS separates ideal from detailed models. Hatched means not simulated. One sentence on top says what the control receives. A click on a chip opens the detail in the inspector.

The control receives V_LOW1, I_LOW1 and V_HIGH every 100 µs, ideal; it returns one duty for both legs. Simulated with Pulsim 2.0, exact PWM edges.

Power
`switching model`

Measurement
`ideal · no sensor error`

Driver
`PWM + dead time`

Controller
`your control.cpp, compiled`

Auxiliary supply
`not simulated`

LEGEND

- simulated as on the board
- simplified: differs from the board
- not simulated

Board-only code in main.cpp (outside setup()) is listed under the Controller chip.

The canvas tint per tab

Your Q2 idea: the canvas keeps its place and changes background, so you always know which job you are in.

Design	Code	Lab	Simulate
values on each block	file and function per block	probes, live readings, checks	fidelity chips